Who Really Finished Where?

Ranking the 2024-25 URC Season Using Linear Algebra

MA283 Project

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Why I Chose This Topic

I chose this topic because I wanted to do project on sports. I've played sports my whole life and I've always loved rugby. I'm an avid Connacht supporter so I wanted to do a project about them. Connacht play in the United Rugby Championship (URC) with the other Irish teams and teams from, Scotland, Wales, Italy and South Africa and they usually don't do too well. They are very inconsistent and can perform well against good teams and very poorly against bad teams, which doesn't fully tell the whole story. Beating Dragons and beating Leinster both earn you four points, even if one of those wins says a lot more about your team than the other.

It turns out you can and has a proper mathematical name: the dominant eigenvector of a dominance matrix. Once I came across the connection to Google's PageRank algorithm, which uses the same underlying idea to rank webpages by importance, I was convinced this was worth exploring. The question I ended up with was simple: if you ranked the 2024-25 URC table in a way that accounted for who beat, not just that you beat them, would anything change?

The Approach

The starting point was building a 16x16 matrix, one row and column for each team. The idea is straightforward: entry *A[i][j]* records the point margin by which team *i* beat team *j*. If team *i* lost, the entry stays near zero (ε). What you end up with is a grid of numbers that encodes the entire season's results in a single object.

Once the matrix exists, you feed it into a process called power iteration. You start by giving every team the same score, a vector of all ones,

URC league standings [edit]

Pos	Team	Pld	W	D	L	PF	PA	PD	TF	TA	TB	LB	Pts
1	Leinster **(CH)**	18	16	0	2	542	256	+286	79	35	11	1	**76**
2	Bulls **(RU)**	18	14	0	4	542	361	+181	71	44	9	3	**68**
3	Sharks	18	13	0	5	436	402	+34	55	59	7	3	**62**
4	Glasgow Warriors	18	11	0	7	468	327	+141	70	40	10	5	**59**
5	Stormers	18	10	0	8	507	418	+89	66	57	11	4	**55**
6	Munster	18	9	0	9	444	429	+15	67	59	11	4	**51**
7	Edinburgh	18	8	1	9	471	407	+64	66	57	9	6	**49**
8	Scarlets	18	9	1	8	427	382	+45	50	52	6	4	**48**
9	Cardiff	18	8	1	9	409	477	−68	63	65	10	3	**47**
10	Benetton	18	9	1	8	393	478	−85	50	65	7	1	**46**
11	Lions	18	8	0	10	402	440	−38	53	60	5	3	**40**
12	Ospreys	18	7	1	10	437	454	−17	60	63	6	4	**40**
13	Connacht	18	6	0	12	420	472	−52	64	62	9	6	**39**
14	Ulster	18	7	0	11	414	506	−92	59	72	5	5	**38**
15	Zebre Parma	18	5	1	12	302	503	−201	38	72	3	4	**29**
16	Dragons	18	1	0	17	335	637	−302	43	92	1	4	**9**

representing the assumption that we know nothing. Then you repeatedly multiply the matrix by the score vector and renormalise. Each multiplication updates every team's score to reflect the quality of the opponents it beat. After enough iterations (around 40 to 50), the vector stops changing. That final vector is the dominant eigenvector, and its entries give you the ranking.

The reason this works is guaranteed by the Perron-Frobenius theorem which I came across when researching Google's PageRank. It states that any strictly positive matrix has a unique largest real eigenvalue with a corresponding eigenvector of all positive entries. Adding a small constant $\varepsilon = 0.01$ to every matrix entry ensures strict positivity, the same trick used in PageRank's 'damping factor' and means the theorem applies cleanly.

The Python implementation was fairly short. The core of it is just 2 lines:

```
v_new = A @ v
v_new = v_new / np.max(v_new)
```

The first line does the matrix-vector multiplication. The second normalises the result so the largest score stays at 1. Everything else in the code is setup, loading the match results, building the matrix, printing the final table.

Problems I Ran Into

This was the most frustrating part, and probably the most instructive. My original plan was to use a complete set of all 144 URC regular-season match results. I found the data on Wikipedia and on all.rugby, which both list every fixture. I didn't want to manually get all the data from Wikipedia, so I had to make a website scraper to get the data from Wikipedia and formatted it into a JSON file as shown.


```
{
  "round": 17,
  "date": "9 May 2025",
  "home_team": "Sharks",
  "home_score": 29,
  "away_score": 10,
  "away_team": "Ospreys",
  "home_bp": true,
  "away_bp": false,
  "venue": "Kings Park Stadium , Durban"
},
```

Findings and Conclusion

The code converged in 21 iterations. Here are the results – the official standings alongside the eigenvector ranking and each team's normalised score:

```
Converged after 21 iterations.

 Off  Team                Pts   Score  Eigen  Shift
-----------------------------------------------------
   1  Leinster             76   1.000      1  -
   2  Bulls                68   0.530      5  dn 3
   3  Sharks               62   0.464      8  dn 5
   4  Glasgow Warriors     59   0.756      2  up 2
   5  Stormers             55   0.664      3  up 2
   6  Munster              51   0.383     11  dn 5
   7  Edinburgh            49   0.537      4  up 3
   8  Scarlets             48   0.405     10  dn 2
   9  Cardiff              47   0.189     13  dn 4
  10  Benetton             46   0.526      6  up 4
  11  Lions                40   0.468      7  up 4
  12  Ospreys              40   0.452      9  up 3
  13  Connacht             39   0.116     14  dn 1
  14  Ulster               38   0.200     12  up 2
  15  Zebre Parma          29   0.107     15  -
  16  Dragons               9   0.012     16  -
PS C:\Users\liamm>
```

The most striking results are in the middle of the table. **Bulls drop from 2nd to 5th** despite 14 wins. Looking at their results, many of their biggest victories came against the weaker teams. They bet Dragons by 40, Ulster by 26 and Zebre by 26. The teams they beat didn't themselves have high eigenvector scores, so those wins don't carry as much weight in the quality-adjusted ranking.

Sharks drop five places from 3rd to 8th, the biggest fall in the table. Their 13 wins look impressive but a lot came in the Welsh pool against sides that rank poorly in the eigenvector.

Glasgow Warriors rise two places from 4th to 2nd with a score of 0.756, second only to Leinster. They beat strong opposition by large margins, notably Cardiff 52-36 away and Dragons 45-20. Their wins were against teams that themselves performed well in the eigenvector.

Munster's drop from 6th to 11th is one of the more surprising results. They finished the season with 9 wins, the same as Scarlets and Benetton, but their eigenvector score of 0.383 is noticeably lower. A lot of their wins came against the lower-ronked teams, and they lost to most of the sides that rank highly in the eigenvector.

Last but not least, my team, **Connacht, performed even worse, going from 13th to 14th**. This is obviously personally disappointing but it makes sense seeing as they only had 6 wins and their margins were very small reducing their eigenvector score.

If I Were to do it Again

A few things that stand out now that I'm on the other side of it.

I'd think harder earlier about what to put in the matrix. I only used point margin's but would love to have tried different options like implementing the URC's own bonus-point to the matrix somehow. It could give different rankings depending on how you apply it. I also could've added home vs away fixtures into the matrix to see if having a home advantage changes any of the eigenvectors.

I would also love to compare other rugby leagues such as the English Premiership or the French Top14 and see if there are many differences with their eigenvalues. Are they more like the league standings or do they differ more.

Sources/References

Bryan, K. & Leise, T. (2006). The $25,000,000,000 eigenvector: the linear algebra behind Google

Cornell University — Mathematics of Google Search

2024–25 United Rugby Championship results and standings. Wikipedia.

Website Scraper tutorial: https://www.youtube.com/watch?v=ng2o98k983k